UNIT – IV

FUNDAMENTALS OF INVENTORY CONTROL

Inventory generally refers to the materials in stock. It is also called as the idle resource of an enterprise. Inventories represent those items which are either stocked for sale or they are in the process of manufacturing or they are in the form of materials, which are yet to be utilized. The interval between receiving the purchased parts and transforming them into final products varies from industries to industries depending upon the cycle time of manufacture. It is, therefore, necessary to hold inventories of various kinds to act as a buffer between supply and demand for efficient operation of the system. Thus, an effective control on inventory is a must for smooth and efficient running of the production cycle with least interruptions.

Meaning of Inventory Control

Inventory control is a planned approach of determining what to order, when to order and how much to order and how much to stock so that costs associated with buying and storing are optimal without interrupting production and sales.

Inventory control basically deals with two problems:

1.When should an order be placed? (Order level)

2. How much should be ordered? (Order quantity)

These questions are answered by the use of inventory models. The scientific inventory control system strikes the balance between the loss due to non-availability of an item and cost of carrying the stock of an item. Scientific inventory control aims at maintaining optimum level of stock of goods required by the company at minimum cost to the company.

Reasons for Keeping Inventories

To stabilize production The demand for an item fluctuates due to a no of factors. Example - seasonality, production schedule etc. The inventories (raw materials and components) should be made available to the production as per the demand. If inventories are not available, it results in stock out and the production stoppage takes place due to lack of materials. Hence, the inventory should take care of the fluctuations so that the production runs smoothly.

To take advantage of price discounts Usually the manufacturers offer discount for bulk buying and to gain this price advantage the materials are bought in bulk even though it is not required immediately. Thus, inventory is maintained to gain economy in purchasing.

To meet the demand during the replenishment period The lead time for procurement of materials depends upon many factors like location of the source, demand supply condition, etc. So, inventory is maintained to meet the demand during the procurement (replenishment) period.

To prevent loss of orders (sales) In this competitive scenario, one has to meet the delivery schedules at 100 per cent service level, means they cannot afford to miss the delivery schedule which may result in loss of sales. To avoid this organizations, have to maintain inventory.

To keep pace with changing market conditions The organizations have to anticipate the changing market sentiments and they have to stock materials in anticipation of non- availability of materials or sudden increase in prices. Sometimes the organizations have to stock materials due to other reasons like supplier's minimum quantity condition, seasonal availability of materials or sudden increase in prices.

Manufacturing Inventory Vs Distribution Inventory

Manufacturing inventory, or production inventory, is all of the supplies and materials on hand meant for the manufacturing of products. Retailers and wholesalers have inventories that include only items ready to sell. But a manufacturing company's inventory consists of goods in multiple stages of production. From raw materials to finished products ready to ship.

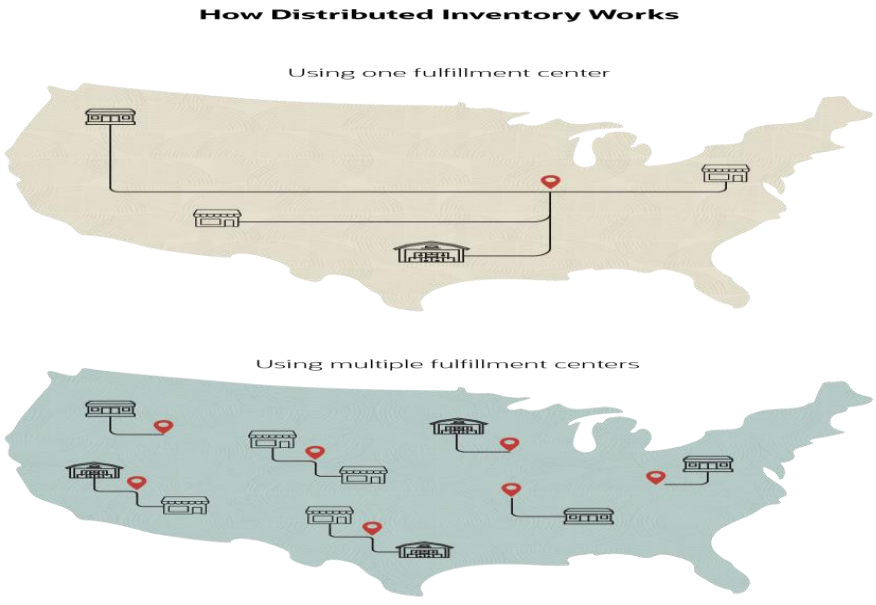
A manufacturing company's balance sheet has three categories for its inventory: raw materials, work-in-process, and finished goods inventory

Raw materials inventory is all the most basic materials needed in production inventory management. Metals, plastic, stone, chemicals, and everything else that gets worked into finished goods.

For example, a company that produces jellies and jams. Some of the primary raw materials would be fruit and sugar (for the jam) and glass (for the jars).

Work-In-Process inventory, is whatever inventory is past the raw material stage but not yet a finished product.

Finished goods are the end of the line. They made it all the way through production and are ready to leave the nest. When a product is ready for sale and shipment, it's a finished good.



Distributed inventory is a goods distribution strategy in which inventory is stored in and shipped from multiple centres or warehouses located in different geographic areas.

In a traditional inventory model, a manufacturer stores its goods in a centralized warehouse. When a customer places an order, the order is prepared for delivery at the central location and is then sent to the customer.

In a distributed inventory model, when a customer places an order, the warehouse or fulfilment centre that is closest to that customer and has all items in stock or that can fulfil it at the lowest cost will prepare and ship it.

Multi-stage Inventory Vs Multi Echelon Inventory

In a multi-stage inventory system, raw materials, work-in-process, and finished goods move through a set of stages to meet the demand.

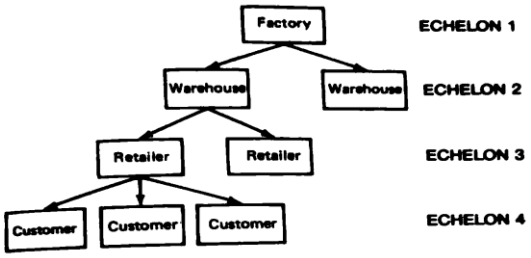


FIGURE [redacted] Multiechelon inventory systems

When parts (which eventually become finished goods) are stocked at more than one point in the sequential production process, there are multistage inventories. These partially finished parts waiting for completion and eventual sales are called as work in process/ work in progress/ goods in process/ in process inventories and if they are stocked at multiple stages in the production process it is called as Multistage inventories.

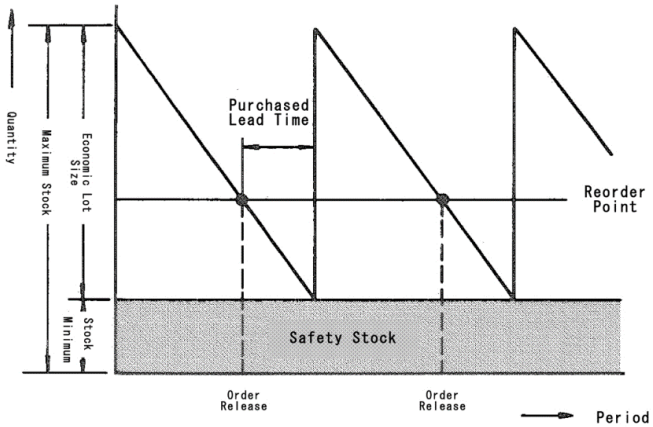
Multi Echelon inventories. As shown in the figure, these are inventories of products at the various levels or echelons – factory, warehouse, retailer, and customer – in a distribution system.

Inventory Systems/ Inventory Control Systems

Inventory control systems are the ordering and monitoring techniques used to control the quantity and timing of inventory transactions. The traditional inventory control systems are classified as either perpetual (continuously monitored) or periodic and numerous combinations of them. In addition, many broad-based scheduling systems incorporate inventory control activities as an integral part of the larger systems.

Q/R (or) Perpetual (or) Fixed Order-Quantity Inventory System

The fixed order quantity system is also known as the Q system. In this system, whenever the stock on hand reaches the reorder point, a fixed quantity of materials is ordered. The fixed quantity of material ordered each time is actually the economic order quantity (EOQ is a technique for inventory management that minimizes ordering and holding costs for the year).

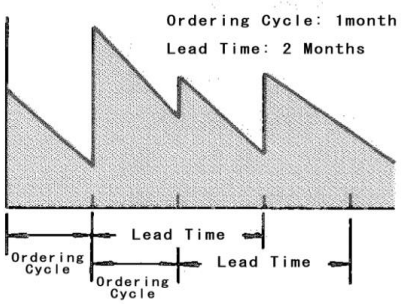


Whenever a new consignment arrives, the total stock is maintained within the maximum and the minimum limits. The fixed order quantity method is a method that facilitates for a predetermined amount of a given material to be

ordered at a particular period of time. This method helps to limit reorder mistakes, conserve space for the storage of the finished goods, and block those unnecessary expenditures that would tie up funds that could be better utilized elsewhere. The fixed order quantity may be bridged to an automatic reorder point where a particular quantity of a good is ordered when stock at hand reaches a level which is already determined.

Fixed Order Period System (P System)

In this system, the stock position of each material of a product is checked at regular intervals of time period. When the stock level of a given product is not sufficient to sustain the operation of production until the next scheduled tested, an order is placed destroying the supply. The frequency of reviews varies from organization to organization. It also varies among products within the same organization, depending upon the importance of the product, predetermined production schedules, market conditions and so forth. The order quantities vary for different materials.



Order Quantity = Scheduled Demand of (Ordering Cycle + Lead Time) + Safety Stock − Inventory Quantity − Released Order

Just-in-Time (JIT) Inventory System

The just-in-time (JIT) inventory system is a management strategy that aligns raw-material orders from suppliers directly with production schedules. Companies employ this inventory strategy to increase efficiency and decrease waste by receiving goods only as they need them for the production process, which reduces inventory costs. This method requires producers to forecast demand accurately.

The just-in-time (JIT) inventory system minimizes inventory and increases efficiency. JIT production systems cut inventory costs because manufacturers receive materials and parts as needed for production and do not have to pay storage costs. Manufacturers are also not left with unwanted inventory if an order is cancelled or not fulfilled.

JIT inventory systems have several advantages over traditional models. Production runs are short, which means that manufacturers can quickly move from one product to another. Also, this method reduces costs by minimizing warehouse needs. Companies also spend less money on raw materials because they buy just enough resources to make the ordered products and no more.

The disadvantages of JIT inventory systems involve potential disruptions in the supply chain. If a raw-materials supplier has a breakdown and cannot deliver the goods promptly, this could conceivably stall the entire production line. A sudden unexpected order for goods may delay the delivery of finished products to end clients.

Inventory Costs

Inventory costs are the costs related to storing and maintaining a business inventory over a certain period of time. Ordering, holding, carrying, shortage and spoilage costs make up some of the main categories of inventory-related costs.

1. Ordering Costs

Ordering costs include payroll taxes, benefits and the wages of the procurement department, labour costs etc. These costs are typically included in an overhead cost pool and allocated to the number of units produced in each period.

- Transportation costs
- Cost of finding suppliers and expediting orders
- Receiving costs
- Clerical costs of preparing purchase orders
- Cost of electronic data interchange

2. Inventory Holding Costs

This is simply the amount of rent a business pays for the storage area where they hold the inventory. This can be either the direct rent which the company pays for all the warehouses or a percentage of the total rent of the office area utilized for storing inventory.

- Inventory services costs
- Inventory risk costs
- Opportunity cost - money invested in inventory
- Storage space costs
- Inventory financing costs

3. Shortage Costs

Shortage costs, also known as stock-out costs, occurs when businesses become out of stock for various reasons. Some of the reasons might be as below:

- Emergency shipments costs
- Disrupted production costs
- Customer loyalty and reputation

4. Spoilage Costs

Perishable inventory stock can rot or spoil if not sold in time, so controlling inventory to prevent spoilage is essential. Products that expire are a concern for many Industries such as the food, beverages. The pharmaceutical, healthcare and cosmetic industries are affected by the expiration and use-by dates of their products.

5. Inventory Carrying Costs

This is the lesser-known aspect of inventory cost. This cost requires a certain amount of calculation to understand the extent of its impact on your Profit & Loss statement. Inventory carrying costs refers to the amount of interest a business loses out on the unsold stock value lying in the warehouses.

Techniques of Inventory Control

In any organization, depending on the type of business, inventory is maintained. When the number of items in inventory is large and then large amount of money is needed to create such inventory, it becomes the concern of the management to have a proper control over its ordering, procurement, maintenance and consumption. The control can be for order quality and order frequency. The different techniques of inventory control are: (1) ABC analysis, (2) HML analysis, (3) VED analysis, (4) FSN analysis, (5) SDE analysis, (6) GOLF analysis and (7) SOS analysis.

The most widely used method of inventory control is known as ABC analysis. In this technique, the total inventory is categorised into three sub-heads and then proper exercise is exercised for each sub-head.

ABC analysis: In this analysis, the classification of existing inventory is based on annual consumption and the annual value of the items. Hence, we obtain the quantity of inventory item consumed during the year and multiply it by unit cost to obtain annual usage cost. The items are then arranged in the descending order of such annual usage cost. The analysis is carried out by drawing a graph based on the cumulative number of items and cumulative usage of consumption cost. Classification is done as follows:

Once ABC classification has been achieved, the policy control can be formulated as follows:

A-Item: Very tight control, the items being of high value. The control need be exercised at higher level of authority.

B-Item: Moderate control, the items being of moderate value. The control need be exercised at middle level of authority.

C-Item: The items being of low value, the control can be exercised at gross root level of authority, *i.e.*, by respective user department managers.

VED analysis: VED analysis is an inventory management technique that classifies inventory based on its functional importance. It categorizes stock under three heads based on its importance and necessity for an organization for production or any of its other activities. VED analysis stands for Vital, Essential, and Desirable.

V-Vital category

As the name suggests, the category "Vital" includes inventory, which is necessary for production or any other process in an organization. The shortage of items under this category can severely hamper or disrupt the proper functioning of operations. Hence, continuous checking, evaluation, and replenishment happen for such stocks. If any of such inventories are unavailable, the entire production chain may stop. Also, a missing essential component may be of need at the time of a breakdown. Therefore, the order for such inventory should be beforehand. Proper checks should be put in place by the management to ensure the continuous availability of items under the "vital" category.

E- Essential category

The essential category includes inventory, which is next to being vital. These, too, are very important for any organization because they may lead to a stoppage of production or hamper some other process. But the loss due to their unavailability may be temporary, or it might be possible to repair the stock item

or part. The management should also ensure optimum availability and maintenance of inventory under the “Essential” category. The unavailability of inventory under this category should not cause any stoppage or delays.

D- Desirable category

The desirable category of inventory is the least important among the three, and their unavailability may result in minor stoppages in production or other processes. Moreover, the easy replenishment of such shortages is possible in a short duration of time.